17 PERMUTATION & COMBINATION

Permutations

The arrangements made by taking some or all elements out of a number of things is called a permutation. Permutation implies "arrangement" where "order of the things" is important.

The permutations of three things, *a, b* and *c* taken two at a time are *ab, ba, ac, ca, cb* and *bc*. Since the order in which the things are taken is important, *ab* and *ba* are counted as two different arrangements.

The number of permutations of *n* things taking *r* at a time is denoted by ${}^nP_r$.

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

Combinations

The groups of selections made by taking some or all elements out of a number of things is called a combination. In combination, the order in which the things are taken is not important.

The combination of three things, *a, b* and *c*, taken two at a time are *ab, bc* and *ca*. Here, *ab* and *ba* are same because the order in which *a* and *b* are taken is not important. The words "combination" and "selection" can be used without any differentiation.

The number of combinations of *n* things taking *r* at a time is denoted by ${}^nC_r$.

$$ {}^nC_r = \frac{n!}{r!(n-r)!} $$

SOME IMPORTANT FACTS

(i) ${}^nC_0 = 1 = {}^nC_n$

(ii) ${}^nP_r = \frac{n!}{(n-r)!}$

${}^nP_r$ = Arrangement of *r* things means first selecting *r* things and arranging these.

(iii) ${}^nC_{r-1} + {}^nC_r = {}^{n+1}C_r$

(iv) ${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots\ldots\ldots + {}^nC_n = 2^n$

(v) ${}^nC_r = \frac{{}^nP_r}{r!}$

(vi) ${}^nC_r = {}^nC_{n-r}$

Permutations and Combinations

with and without repetitions.

(i) Permutation with repetition = ${}^{n+r-1}P_r$

(ii) Permutation without repetition = ${}^nP_r$

(iii) Combination with repetition = ${}^{n+r-1}C_r$

(iv) Combination without repetition = ${}^nC_r$

FUNDAMENTAL RULE OF COUNTING

I. Product Rule (Multiplication theorem)

If there are *m* ways of doing a first things and *n* ways of doing a second thing, then the total number of ways of doing these things is $m \times n$.

e.g. In a class there are 9 boys and 8 girls. The teacher wants to select a boy and a girl to represent the class in a function. In how many ways can the teacher make this selection ?

Sol. : Total no. of ways = 9 × 8 = 72.

II. Sum Rule (Addition theorem)

If a work *w* consists of two part w_1 and w_2 of which one part can be done in *m* ways and other can be done in *n* ways, then the work *w* can be done in (m + n) ways.

e.g. In a class there are 9 boys and 8 girls. The teacher wants to select either a boy or a girl to represent the class in a function. In how many ways can the teacher make this selection ?

Sol. : Total no. of ways = 9 + 8 = 17.

Summary of Laws

If you find that there is 'AND', then you have to multiply (×) the cases.

If you find that there is 'OR', then you have to add (+) the cases.

Circular Permutation

Number of circular permutation of *n* different things taken all at a time = (n – 1) !

In a case of a necklace or garland as anticlockwise and clockwise arrangements are same. So, total number of arrangements of *n* beads for forming a necklace = $\frac{1}{2}(n-1)!$

SOME MORE IMPORTANT POINTS

1. Number of permutations of *n* different things taken *r* at a time when repetition is allowed = n nnn……r times = n^r.

2. Permutation of Things when some are identical : If out of *n* things, *p* are exactly alike of one kind, *q* exactly alike of second kind and r exactly alike of third kind and the rest are different, then the number of permutations of *n* things taken all at a time a = $\frac{n!}{p! \times q! \times r!}$

3. If n articles are arranged in such a way that 2 particular articles never come together. The number of such arrangement is (n-2) (n-1)!

SOLVED EXAMPLES

Ex.1 How many numbers are there between 100 and 1000. Such that every digit is either 3 or 4.

⇒ 2 × 2 × 2 = 8

Ex.2 8 examination papers are arranged in such a way that best and worst papers never come together. Find the number of arrangements.

⇒ No. of arrangements = (n – 2) (n – 1)! = 6 × 7!

Ex.3 In how many ways 5 men and 4 women can be seated at a round table. If

i. All the four women sit together

ii. All the four women do not sit together.

⇒ i. Assuming all the women to be one group, we have 6 things (5 men + 1 group) to be arranged around a circular table which can be arranged in 5! ways However, these 4 women can be arranged in 4! Ways among themselves.

So, the total number of arrangements = 5! × 4!

ii. The total number of arrangements when there is no restriction = 8! and the number of arrangements when all the four women sit together = 5! ×4!. So, the total number of arrangements when all the four women do not sit together = 8! – 5! ×4!

Ex.4 Find the value of r if ${}^{16}C_r = {}^{16}C_{r+2}$

⇒ We know that,

${}^{n}C_r = {}^{n}C_{n-r} \Rightarrow {}^{16}C_r = {}^{16}C_{16-r} \Rightarrow {}^{16}C_{16-r} = {}^{16}C_{r+2}$

$\Rightarrow 16 - r = r + 2 \Rightarrow 2r = 14 \Rightarrow r = 7.$

Ex.5 Out of 9 consonants and 5 vowels, how many words can be formed each containing 5 consonants and 3 vowels.

⇒ 5 consonants can be chosen out of 9 in ${}^{9}C_5$ and 3 vowels can be chosen out of 5 in ${}^{5}C_3$ ways. So the number of selection having 5 consonants and 3 vowels = ${}^{9}C_5 \times {}^{5}C_3$

Each of these selections contains 8 letters which can be arranged among themselves in 8! Ways.

Total number of words = ${}^{9}C_5 \times {}^{5}C_3 \times 8!$

EXERCISE

1. In how many ways can the letters of the word "SUMMER" be rearranged ?

(a) 560 (b) 360 (c) 720 (d) 1260

2. A student is to answer 10 out of 12 questions in an examination such that he must choose at least 3 from the first four questions. The number of choices available to him is

(a) 140 (b) 56 (c) 32 (d) 42

3. Find the number of ways in which 13 different beads can be arranged to form a necklace.

(a) 13! (b) 2 x 12!

(c) $\frac{1}{2} \times 12!$ (d) None of these

4. In how many ways can the letters of the word PENCIL be arranged so that P and C are next to each other ?

(a) 118 (b) 116 (c) 240 (d) 120

5. In how many ways can 15 billiard balls be arranged in a row if 6 are black, 5 red and 4 white ?

(a) $\frac{16! \times 6!}{5! \times 4!}$ (b) $\frac{16!}{6! \times 5!}$ (c) $\frac{15!}{6! \times 5! \times 4!}$ (d) $\frac{15!}{5! \times 4!}$

6. How many different 7 digit telephone numbers can be formed from 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

(a) 10^6 (b) 9×10^5 (c) 9×10^6 (d) 10^7

7. How many three digits numbers can be formed without using the digits 2, 3, 5, 7.

(a) 180 (b) 162 (c) 125 (d) 156

8. From 3 officers and 7 supervisor, in how many ways can a team of 5 be chosen to include exactly one officer ?

(a) 330 (b) 225 (c) 224 (d) 105

9. How many two digit odd numbers can be formed from the digits 1, 2, 5, 6 and 7, if the repetition of digits is allowed ?

(a) 12 (b) 15 (c) 18 (d) 22

10. Sita has 5 friends in how many ways can she invite one or more of them at a dinner party.
(a) 63 (b) 31 (c) 68 (d) 73

11. In how many ways can be letters of the words "DISCOVERY" be arranged so that the vowels may never be separated ?
(a) 6! × (3)! (b) 8! × 4! (c) 7! × 3! (d) 8!

12. There are 7 pairs of socks of different size. In how many ways can you choose one for the left foot and one for the right foot such that they are not of the same pair?
(a) 56 (b) 52 (c) 50 (d) 42

13. 7 chairs are numbered from 1 to 7. Two women first sit on two chairs from those marked 1 to 3 and 3 men select 3 chairs from the remaining. Find the number of possible arrangements.
(a) 1250 (b) 560 (c) 360 (d) 1440

14. How many new words can be formed with the word ARRANGEMENT all ending in G?
(a) 10 ! × 2 ! × 2 ! (b) 10 ! × (2 !) / 3
(c) 10 ! × 2 ! (d) $\frac{10!}{2!\times2!\times2!\times2!}$

15. The number of ways of selecting 8 men and 2 women from 13 men and 10 women is :
(a) ${}^{23}C_{10}$ (b) ${}^{13}C_8 \times {}^{10}C_2$
(c) ${}^{13}C_2 \times {}^{10}C_8$ (d) None of these

16. How many seven letter words can be formed from the letters of the word "COUNTRY" beginning with C and ending with Y ?
(a) 120 (b) 750 (c) 5040 (d) 720

17. After a get together every person present handshakes with every other person. If there were 91 hands shakes in all, how many persons were present in the party ?
(a) 14 (b) 13 (c) 15 (d) 16

18. In a cricket championship, there were 120 matches played. Every two teams played one match with each other. The number of teams participating in the championship is
(a) 18 (b) 19 (c) 17 (d) 16

19. A women wants to select 3 toys for his child. I variety of toys has 8 models, II variety has 5 models and III variety has 3 models. In how many ways can he select the 3 toys, one from each of the variety?
(a) 16 (b) 119 (c) 120 (d) 108

20. In how many ways can 6 physics, 5 chemistry and 4 math books be arranged, if all the books of each type are to be kept together ?
(a) ${}^{15}P_6 \times {}^{9}P_4 \times {}^{5}P_5$ (b) ${}^{15}P_{15}$
(c) 12441600 (d) 6930

21. There are three prizes to be distributed among six students. If no students get more than one prize then find the no. of ways:
(a) 10 ways (b) 20 ways
(c) 60 ways (d) 80 ways

22. A class photograph has to be taken. The front row consists of 5 girls who are sitting. 18 boys are standing behind. The two corner positions are reserved for the 2 tallest boys. In how many ways can the students be arranged ?
(a) 18 ! × 240 (b) 16 ! × 240
(c) 16 ! × 2! × 240 (d) None of these

23. The number of ways in which 4 men and 4 women can sit at a round table if no two women are to sit together is given by :
(a) 6 ! × 4 ! (b) 4 ! × 3 !
(c) 20 (d) 6 ! × 5 !

24. In how many ways can 6 persons be seated around a circular table such that two particular persons are next to each other ?
(a) 24 (b) 48 (c) 56 (d) 72

25. How many words can be formed from the letters of the word 'EXTRA' so that the vowels are never together ?
(a) 100 (b) 320
(c) 426 (d) 72

26. A candidate is required to answer 6 out of 10 questions divided into 2 groups each containing 5 questions. He can attempt maximum 4 questions from each group. In how many ways can he make up his choice ?
(a) 660 (b) 260
(c) 200 (d) 566

27. Out of 7 men and 4 women a committee of 4 is to be formed. Find the number of ways in which this can be done so as to include atleast 2 women.
(a) 175 (b) 165
(c) 155 (d) 321

28. In how many ways can 6 boys and 6 girls sit down at a round table in such a way that no 2 boy being in the consecutive positions ?
(a) 7 ! × 6 ! (b) 6 ! × 5 !
(c) 3 ! × 4 ! (d) 5 ! × 4 !

29. Ten different letters of an alphabet are given, words with 4 letters are formed from these given letter. Then, the number of words which have at least one letter repeated is
(a) 10000 (b) 5040
(c) 4960 (d) None of these

30. In how many ways can 4 cards be selected from a pack of cards as to include atleast one club ?
(a) ${}^{52}C_4 - {}^{4}C_4$ (b) ${}^{39}C_4$
(c) ${}^{52}C_4$ (d) ${}^{52}C_4 - {}^{39}C_4$

31. Sush went to a beach with her friends. Before arriving at the beach sush entered into a quarrel with one of her friends. After the quarrel they decided that they would never stand or sit adjacent to each other. Keeping this in mind one of her friends realized that if they were to sit in a row at the beach they could still sit in 480 ways. How many friends had accompanied Sush to the beach?
(a) 4 (b) 6 (c) 5 (d) 7

32. A committee of 8 persons has to be formed from 15 persons A, B, C, D …….. O. In how many ways is it possible if, A, E, I are always together but B, H and K will not be there if E is in the committee? Also, F and G are always selected.
(a) 45 (b) 245 (c) 119 (d) None

33. A series is formed as: 1, 4, 7, 10, and 13 …… 97. Two integers are chosen from this series. What is the probability that sum of these two integers is odd?
(a) $\frac{17}{32}$ (b) $\frac{17}{33}$ (c) $\frac{17}{31}$ (d) None

34. The coach of the Indian cricket team has to select and fit in 11 players out of Rahul, Pathan, Sachin, Dhoni, Yuvraj, Anil, Gautam, Ajit, Jay, Sourav, Kaif, Amol, Wasim and Laxman in 11 batting slots (numbered 1, 2, ……., 11). If it is certain that Sachin will be picked and Sourav will not be picked, in how many ways can the coach set a batting line-up?
(a) 66 × 11! (b) ${}^{12}P_{10}$
(c) ${}^{13}P_{11}$ (d) 65 × 11!

35. At the beginning of a party, each person present shook hands with all other people present and there were in all 28 handshakes. In the midst of the party, 2 persons left due to an emergency. Now, the number of males and females present in the party was equal. At the end, each female shook hands only with every female present and each male shook hands only with every male present. What is the total number of handshakes that took place at the party?
(a) 52 (b) 45 (c) 35 (d) 34

36. In how many ways can your arrange the letters 'APTITUDE' so as to form words that begin and end with a vowel?
(a) 6400 (b) 1280 (c) 4326 (d) 4320

37. In how many ways can you arrange 4 blocks of the same shape and size in a row from left to right after selecting them from one orange, one green, one black, one purple, one white, two red, two yellow and two blue blocks?
(a) 2454 (b) 1950 (c) 990 (d) 975

38. There are 9 books; 2 by Sidney, 3 by Archer and the rest by John. These books are to be arranged linearly on a shelf such that:
i. no 2 books by John are adjacent.
ii. at least one book by Archer is at an end.
iii. except for John and Archer, the books by the same author are indistinguishable. Find out the possible number of arrangements.
(a) 3450 (b) 8208 (c) 2595 (d) 1290

39. It is known that the number of ways of selecting a team of x students out of 5 students is same as the number of ways of not selecting those x students and selecting all the rest. If the sum of these two number of ways is 20, x is :
(a) 2 (b) 3 (c) 2 or 3 (d) None

40. Set S contains three distinct positive integers and their respective successors. Also, no three elements in S are consecutive. Find the number of non-empty subsets of S that contain no two consecutive numbers.
(a) 36 (b) 26 (c) 77 (d) 48

41. There are 6 tasks and 6 persons. Task 1 cannot be assigned either to person 1 or to person 2; task 2 must be assigned to either person 3 or person 4. Every person is to be assigned one task. In how many ways can the assignment be done?
(a) 144 (b) 189 (c) 192 (d) 368

42. Let S be the set of five-digit numbers formed by the digits 1, 2, 3, 4 and 5, using each digit exactly once such that exactly two odd positions are occupied by odd digits. What is the sum of the digits in the rightmost positions of the numbers in S?
(a) 228 (b) 194 (c) 294 (d) 216

43. N persons stand on the circumference of a circle at distinct points. Each possible pair of persons, not standing next to each other, sings a two –minute song one pair after the other. If the total time taken for singing is 28 minutes, what is N?
(a) 7 (b) 5 (c) 3 (d) None

44. There are 12 towns grouped into four zones with three towns per zone. It is intended to connect the towns with telephone lines such that every two towns are connected with three direct lines if they belong to the same zone, and with only one direct line otherwise. How many direct telephone lines are required?
(a) 72 (b) 96 (c) 90 (d) 140

45. There are 6 boxes numbered 1, 2, … 6. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is
(a) 21 (b) 25 (c) 33 (d) 60

45.(c) Let $S = \frac{1}{6}+\frac{5}{36}+\frac{19}{216}+\frac{65}{1296}+\frac{211}{7776}+\ldots\ldots\ldots$

$$= \frac{3^1-2^1}{6^1}+\frac{3^2-2^2}{6^2}+\frac{3^3-2^3}{6^3}+\frac{3^4-2^4}{6^4}+\ldots\ldots$$

$$= \left(\frac{1}{2^1}-\frac{1}{3^1}\right)+\left(\frac{1}{2^2}-\frac{1}{3^2}\right)+\left(\frac{1}{2^3}-\frac{1}{3^3}\right)+\ldots\ldots\ldots$$

$$= \left(\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\ldots\ldots\right)-\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\ldots\right)$$

Thus, the given series is the difference of two infinite geometric series with their respective sums S_1 and S_2 given by : $s_1 = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$ and $s_2 = \frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{1}{2}$

$\therefore$ $S = 1 - \frac{1}{2} = \frac{1}{2}$. Hence (c).

17 PERMUTATIONS & COMBINATIONS

1.(b) Total number of arrangements $= \frac{6!}{2!} = 360$

2.(c) Required no. of ways $= {}^4C_3 \times {}^8C_7 = 32$

3.(c) 13 different beads can be arranged among distinction between clockwise arrangements and anticlockwise arrangements. So, the required number of arrangements

$= \frac{1}{2}\times 12!$

4.(c) 5! x 2! = 240

5.(c) Required number of ways $= \frac{15!}{6!\times 5!\times 4!}$

6.(c) 0 can not take the first place, so there are 9 possibilities for the first place. For the remaining 6 places, the 10 digits can appear at any of the places in 10^6 ways.
So total no. of ways $= 9 \times 10^6$

7.(a) Units and ten's place can be filled in 6 ways. Hundred's place can be filled in 5 ways (0 cannot be taken in hundred's place). Hence, required number of ways = 6×65 = 180)

8.(d) One officer and 4 supervisor can be chosen in ${}^3C_1 \times {}^7C_4 = 105$ ways

9.(b) The number is odd if 1 or 5 or 7 appears in the unit's place. Therefore, there are three ways of filling the unit place. Since, the repetition of digits is allowed, the ten's place can be filled by any of the five digits 1, 2, 5, 6 and 7.
Hence, number of two digits odd numbers
= 3 × 5 = 15.

10. (b) Total ways $= 2^n - 1 = 2^5 - 1 = 32 - 1 = 31$.

11.(c) Considering all vowels *E, I, O* as one unit the seven letters can be arranged in ${}^7P_7 = 7!$ ways
The three vowels can be arranged themselves in ${}^3P_3$ = 3! ways
Total number of arrangements = 7 ! × 3 !

12.(d) Number of possible arrangements
$= {}^7C_1 \times {}^6C_1 = 7\times 6 = 42$

13.(c) Total number of arrangements $= {}^3P_2 \times {}^5P_3 = 360$

14.(d) Required number of arrangements $= \frac{10!}{2!\times 2!\times 2!\times 2!}$

15.(b) The no of ways $= {}^{13}C_8 \times {}^{10}C_2$

16.(a) The first place will always be taken by C and last place always be taken by Y. The remaining five places can be filled by the remaining five letters in ${}^5P_5$ ways.
$\therefore$ Total numbers of words = 1×1 = 5 ! = 120

17.(a) Let the total number of persons present in the party be x,

Then, $\frac{x(x-1)}{2} = 91 \Rightarrow x = 14$

18.(d) Let there were x teams participating in the Championship, then total number of matches
$= {}^nC_2 = 120$.

$$\frac{n!}{2!(n-2)!} = \frac{1}{2}(n-1)\times n = 120$$

$\Rightarrow n^2 - n - 240 = 0 \Rightarrow n^2 - 16n + 15n - 240 = 0$
$\Rightarrow n(n - 16) + 15(n - 16) = 0 \Rightarrow (n - 16)(n + 15) = 0$
$\Rightarrow n \neq -15,\quad n = 16$

19.(c) By fundamental principle of counting, required number of ways = 8×5×3 = 120

20.(c) Total number of arrangement = 3! × 6! × 5! × 4! = 12441600

21.(b) 3 prizes among 6 students can be distributed in ${}^6C_3$ ways = 20 ways.

22.(b) Two tallest boys can be arranged in 2 ! ways, rest 16 boys can be arranged in 16! ways.
Girls can be arranged in 5 ! ways.
∴ Total number of ways in which all the students can be arranged = 2! × 16! × 5!
= 16! × 2 × 120 = 16! × 240

23.(b) Required no. of ways = 4 ! × 3 !

24.(b) Consider two particular persons as one person, we have 5 persons in all. These 5 persons can be seated around circular table in (5 - 1)! = 4! ways. But corresponding to these 4! ways, the two particular persons can be arranged among themselves in 2! Ways.
∴ Total number of arrangements = 4! x 2! = 48 ways.

25.(d) The given word contains 5 different letters.
Taking the vowels A, E together, we treat them as one letter
These letters can be arranged in 5 ! = 120 ways
If two vowels are together = 24 way
Number of words, each having vowels together
= (24 × 2) = 48
Required arrangement = 120 – 48 = 72.

26.(c) $2({}^5C_4 \times {}^5C_2) + ({}^5C_3 \times {}^5C_3) = 200$

27.(c) The committee can be formed in the following ways
2 women, 2 men; 3 women, 1 man and 4 women
∴ Required number of ways
$= {}^4C_2 \times {}^7C_2 + {}^4C_3 \times {}^7C_1 + {}^4C_4 \times {}^7C_0 = 155$

28.(b) If position of one boy is fixed, then the remaining 5 can be arranged in 5! Ways = 120 ways
For each such arrangement, there are 6 positions for 6 girls and they can be arranged in 6 ! ways.
Total number of arrangements = 6 ! × 5 !

29.(c) Number of words which have at least one letter repeated
= Total number of words
– total number of words in which no letter is repeated.
$= 10^4 - {}^{10}P_4 = 10000 - 10 \times 9 \times 8 \times 7 = 10000 - 5040 = 4960$

30.(d) Four cards with at least one club can be selected in ${}^{52}C_4 - {}^{39}C_4$ ways.

31.(c) If the group consisted of n person's then possible number of ways when sush does not sit with one particular friend
= n! – 2 × (n – 1)!
Now, n! – 2 × (n – 1)! = 480
⇒ (n – 1)! (n – 2) = 480
= 2 × 3 × 4 × 5 × 4
⇒ (n - 1)! (n – 2) = 5! × 4
Hence there were 6 persons in the group including sush, i.e., she went with five friends.
Hence, (c).

32.(b) If E is in the committee then committee also consists A and I.
B, H and K will not be there in the committee.
Thus, we have to choose 8 – 3 = 5 persons
From 15 – 6 = 9 persons. F and G are always there in the committee.
∴ Number of ways
$= {}^{9-2}C_{5-2} = {}^7C_3 = \frac{5 \times 6 \times 7}{2 \times 3} = 35$ ways.
If E is not there in the committee then there are 15 – 3 = 12 persons to choose from, F and G are there in the committee.
∴ Number of ways $= {}^{12-2}C_{8-2} = {}^{10}C_6 = \frac{7 \times 8 \times 9 \times 10}{1 \times 2 \times 3 \times 4}$
= 210 ways.
Thus, total number of ways = 35 + 210 = 245 ways.
Hence, (b)

33.(b) Number of integers in series be 'n'.
Then, 1 + (n - 1) (3) = 97
∴ (n – 1)3 = 96
∴ n – 1 = 32
∴ n = 33
∴ Of these 33 integers, 16 are even and 17 are odd.
Now, sum of two integers is odd only when one of them is even and other one is odd.
∴ We have to select one even and one odd integer.
∴ Number of ways of selecting the two integers
$= {}^{16}C_1 \times {}^{17}C_1$
∴ The required probability $= \frac{{}^{16}C_1 \times {}^{17}C_1}{{}^{33}C_2}$
$= \frac{16 \times 17 \times 2}{32 \times 33} = \frac{17}{33}$.
Hence, (b)

34.(a) This is a problem of arranging 11 out of 14 players in 11 positions where 1 player is sure to be selected and one is sure not to be selected. Thus effectively the coach has to select only 10 (Sachin being already selected) out of 12 players (excluding Sachin and sourav) and further arrange 11 players in 11 batting slots.
The number of ways
$= {}^{12}C_{10} \times 11! = \frac{12 \times 11}{2} \times 11! = 66 \times 11!$.
Hence (b).

35.(d) Let there be a people in the party.

$^nC_2 = 28$

$$\frac{n!}{(n-1)!\,2!} = 28$$

$\Rightarrow$ n (n-1) = 56

$\therefore$ n = 8

Then, 2 people left the party and from the remaining people, number of males and females is equal. So, there are 3 males and 3 females each.

At the end, people shake hands only with the persons of the same gender.

So, in the end $^3C_2 + {}^3C_2 = 3 + 3 = 6$ handshakes will take place.

$\therefore$ Total number of handshakes = 28 + 6 = 34.

Hence, (d).

36.(d) There are 4 distinct vowels in 'APTITUDE' To begin and end a word with a vowel, the number of possibilities $= {}^4P_2 = 4 \times 3 = 12$

For each such pair of vowels, the other 6 letters can be arranged in:

$\frac{6!}{2!}$ [as there are 2 T's] = 360 ways

$\therefore$ Total number of ways = 12 × 360 = 4320 ways. Hence (d)

37.(a) There are 11 blocks in all.

Four blocks are to be selected.

Now, as there are boxes with repeated colours, we consider the following cases:

Case 1: All the four colours are different.

There are 8 different colours and four out of these can be chosen in $^8C_4$ ways and can be arranged in 4! = 24 ways each.

Total arrangements = $^8C_4 \times 4! = 8 \times 7 \times 6 \times 5 = 1680$

Case 2 : Two blocks of the same colour and the other two of different colours.

The two block of the same colour can be selected in 3 ways and the other two can be selected in $^7C_2$ ways. These 4 can be arranged in 12 ways.

Total arrangements = 3 × 21 × 12 = 756

Case 3: Four blocks in two colours.

The two colours can be selected in 3 ways and the blocks can be arranged in 6 ways. Total arrangements = 3 × 6 = 18

Hence, total number of ways = 1680 + 756 + 18 = 2454. Hence, (a)

38.(b) Archer's books are represented by A, Sydney's by S and John's by J.

We consider 3 cases.

We consider 3 cases:

Case (i):

'A' at both ends.

A J __ J __ J __ J A

2 As can be selected out of 3 in $^3C_2$ ways and can be placed at the end in 2 ways.

The 4 Js can be placed in 4! Ways.

The remaining A and 2 Ss can be placed in 3 ways a the 2 Ss are indistinguishable

$\therefore$ Number of ways = $^3C_2 \times 2 \times 4! \times 3 = 18 \times 24 =$ 432 ways.

Case (ii) :

'A' at one end and 'J' at the other.

The following are the possibilities:

A __ J __ J __ J __ J

A J __ J __ J __ __ J

A J __ __ J __ J __ J

A J __ J __ __ J __ J

Thus, one 'A' can be selected in 3 ways and can be placed at any one of the two ends in 2 ways. The 4 'Js' can be placed in 4 ways as shown and arranged among themselves in 4! Ways.

The remaining 2 As and 2 Ss can be placed in $\frac{4!}{2!} = 12$ ways as the 2 Ss are indistinguishable.

Number of ways = 3 × 2 × 4 × 4! × 12 = 6912.

Case (iii) :

'A' at one end and 'S' at the other.

A J __ J __ J __ J S

A can be selected in 3 ways and placed at the end in 2 ways.

S can be selected in 1 way.

J can be placed in 4! Ways.

The remaining 2 as and 1 S can be placed in 3! = 6 ways.

$\therefore$ Number of ways = 3 × 2 × 4! × 6 = 36 × 24 = 864 ways.

$\therefore$ Total number of ways = 432 + 6912 + 864

= 8208 ways. Hence, (b)

39.(c) Given $^5C_x + {}^5C_{5-x} = 20$ (not selecting 'x' boys is same as selecting the other '5 – x' boys)

Also, $^5C_x + {}^5C_{5-x} = 20$ (given)

$\Rightarrow {}^5C_x + {}^5C_{5-x} = 20$ [$\because {}^nC_r + {}^nC_{n-r}$]

$\Rightarrow {}^5C_x = 10$

But $^5C_x = 10$

$\Rightarrow$ x = 2 or x = 3.

Hence, (c)

40.(b) Let S = {a, b, c, a + 1, b + 1, c + 1}
No three elements in S are consecutive; hence, all the six elements in S are distinct.
The number of singleton subsets = 6
The subsets with exactly two elements, which are consecutive, can be formed by selecting any one of the pairs a, a + 1, b, b + 1 and c, c + 1 in 3 ways.
∴ The number of subsets with exactly two elements, which are non-consecutive = ${}^6C_2 - 3 = 15 - 3 = 12$.
The subsets with exactly three elements and two of them consecutive, can be formed by selecting any one of the pairs, a, a + 1, b, b+ 1 and c, c + 1, in 3 ways and then selecting the third element out of the remaining 4 elements in 4 ways, i.e. in 3× 4 ways.
∴ The number of subsets with exactly three elements and no two consecutive
$= {}^6C_3 - 3 \times 4 = 20 - 12 = 8$.
Any subset of S with more than three elements will contain at least two consecutive elements.
Thus, the total number of subsets of S with no two consecutive elements = 26.
Hence, (b)

41.(a) Task 2 can be assigned to person 3 or 4 in 2 ways.
Task 1 can be assigned to the 3 persons i.e., person 4, 5 or 6 in 3 ways.
Remaining 4 tasks can assigned to remaining 4 persons in 4! Ways = 24 ways.
∴ Required number of ways = 2 × 3 × 24 = 144 ways.
Hence, (a).

42.(d) There are 3 odd places and 3 number for choice
Case 1: First and third places are taken by odd digits

1	___	5	___	2/4	4 cases → sum = 12
5	___	1	___	2/4	4 cases → sum = 12
1	___	3	___	2/4	4 cases → sum = 12
3	___	1	___	2/4	4 cases → sum = 12
5	___	3	___	2/4	4 cases → sum = 12
3	___	5	___	2/4	4 cases → sum = 12

Thus sum of digits in right most positions = 72
Case 2: First and fifth places are taken by odd digits

1	___	___	___	2/4	→ 4 cases → sum = 20
5	___	___	___	2/4	→ 4 cases → sum = 4
1	___	___	___	2/4	→ 4 cases → sum = 12
3	___	___	___	2/4	→ 4 cases → sum = 4
5	___	___	___	2/4	→ 4 cases → sum = 12
3	___	___	___	2/4	→ 4 cases → sum = 20

Thus sum of digits in right most positions = 72
Case 3 : Third and fifth places are taken by odd digits.
As in case 2, sum of digits in the right most positions will be 72.
Hence required sum = 72 × 3 = 216.
Hence, (d)

43.(a) Total time taken for singing is 28 minutes, and each song is 2 minutes long.
∴ 14 songs were sung.
The number of ways in which a pair can be selected such that they are not standing beside each other = (Total number of ways of selecting pairs) – (Number of pairs with persons standing beside each other)
$= {}^NC_2 - N$
$\therefore\ {}^NC_2 - N = 14$

$$\therefore\ \frac{N!}{(N-2)!2!} - N = 14$$

$$\Rightarrow \frac{N(N-1)}{2} - N = 14$$

$\Rightarrow N^2 - N - 2N = 28$
$\Rightarrow (N + 4)(N - 7) = 0$
$\Rightarrow N = -4$ or $N = 7$
Thus total number of people standing on the circumference is 7.
Hence, (a)

44.(c) Let A, B, C, D, be the four zones. Each town in a zone is connected to the other with three direct lines and hence, it requires 9 lines in a zone. There are 4 such zones. Hence, 9 × 4 = 36 direct lines are required.
Each town in A is connected to each town in B, C and D and this requires 3 × 9 = 27 lines.
Now, each town in B is connected to a town in C and D and this requires 3 × 6 = 18 lines.
Also, each town in C is connected to a town in D and this requires 3 × 3 = 9 lines.
Total number of lines required = 36 + 27 + 18 + 9 = 90
Hence, (c).

45.(a) $\overline{1}\ \overline{2}\ \overline{3}\ \overline{4}\ \overline{5}\ \overline{6}$
Since the green balls have to be always kept in consecutive boxes.
One green ball can be arranged in ${}^6C_1 = 6$ ways
two green balls can be arranged in ${}^5C_1 = 5$ ways
three green balls can be arranged in ${}^4C_1 = 4$ ways
Hence, total number of ways

$$= \frac{n(n+1)}{2};\ \text{where } n = 6 = \frac{6 \times 7}{2} = 21.$$

Hence (a).